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Biopsychosocial and Epigenetic Markers of Heat-Stress and Posttraumatic Outcomes in Emergency Responders

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Firefighters showed a greater decline in Hannum age acceleration than paramedics over 12 months.

-2.33 years

01 Study and analysis

Emergency responder professions differ in their exposures. We therefore examined whether epigenetic ageing changed differently by profession.

332 **participants at baseline**

168	99	65
Paramedics	Emergency medical dispatchers	Firefighters

T1 → 12 months → T2

Psychosocial assessments + saliva DNA methylation

Four epigenetic clocks

- Horvath
- Hannum [1]
- PhenoAge
- GrimAge

MethylationEPIC v2 data were analysed with mixed-effects models including Timepoint × Profession and participant random intercepts.

Adjusted for sex, ethnicity, trauma exposure, BMI and estimated cell composition.

02 Hannum change over time

Only Hannum showed evidence that trajectories differed by profession. Dispatchers did not differ significantly from paramedics.

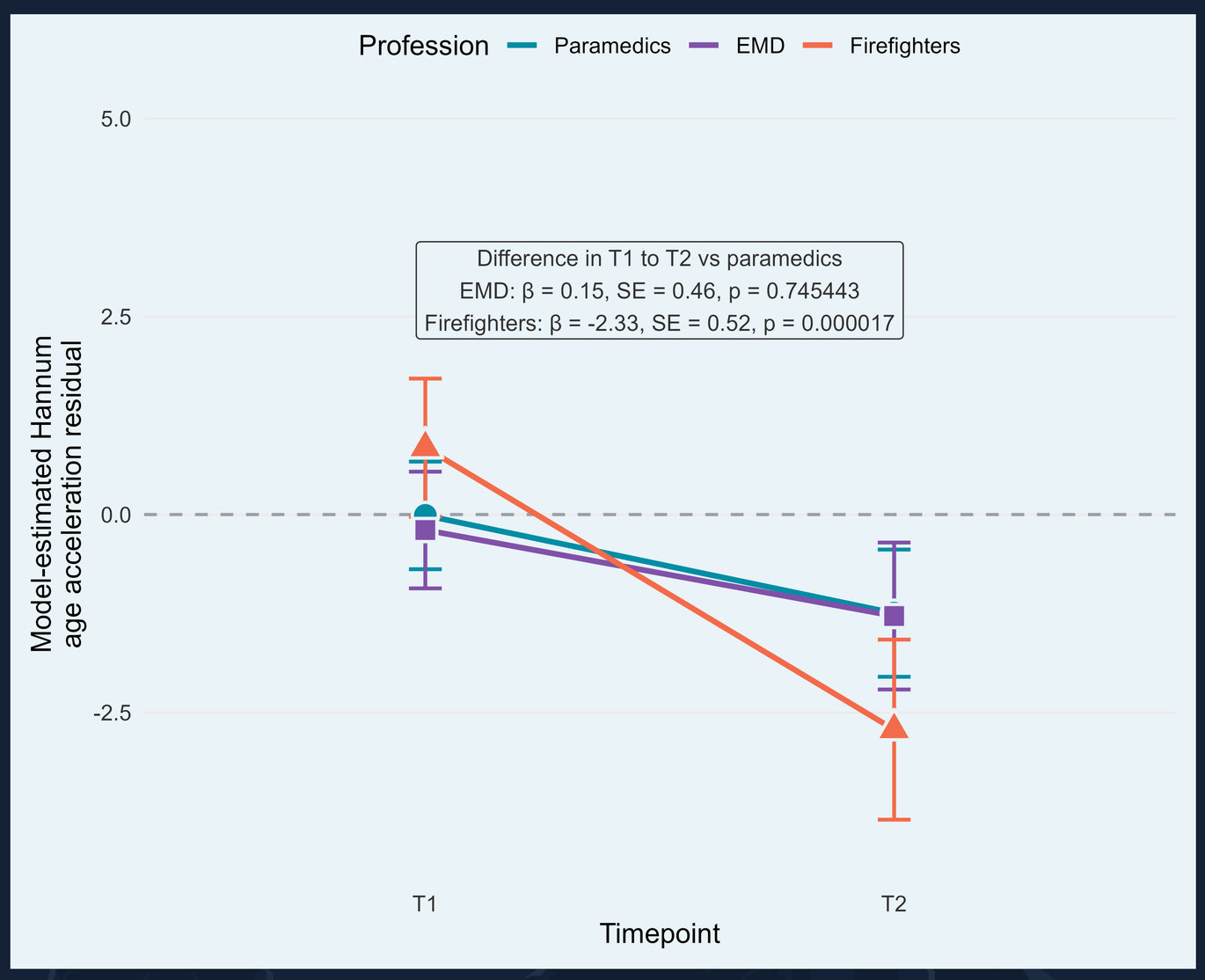


Figure 1. Model-estimated Hannum trajectories. EMD = emergency medical dispatchers. Adjusted analysis: 464 observations from 330 participants; p-values test individual Timepoint × Profession coefficients.

03 Biological processes

We then explored biological processes linked to differences in methylation change over time between firefighters and paramedics.

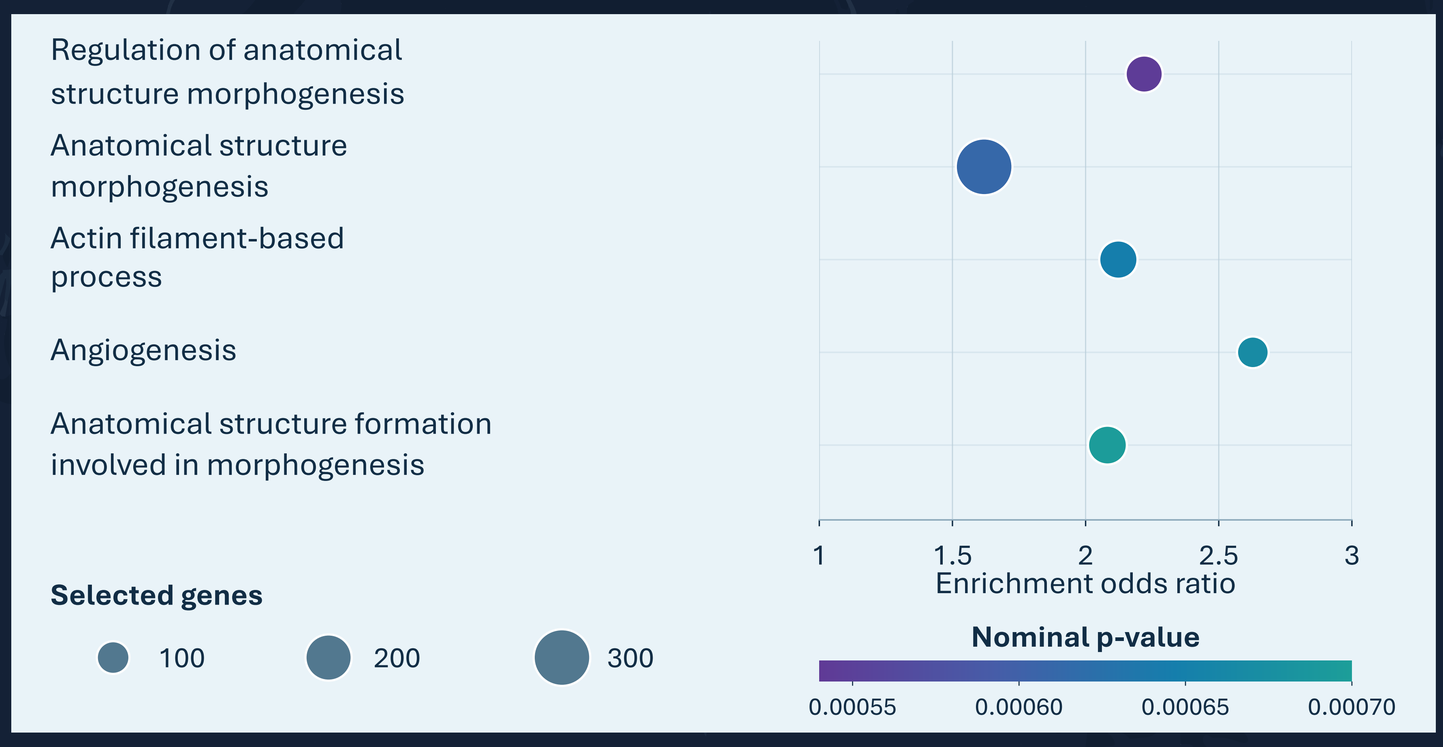


Figure 2. GOstats used a heat-response reference universe and CpGs selected at nominal p < 0.05. Enrichment p-values are unadjusted.

04 Interpretation

Possible immune recovery

Blood studies link Hannum to immune cell composition. Therefore, recovery from recent immune activation is plausible, but was not tested here [4].

Psychological factors

Psychological symptoms and positive factors showed no association across the four clocks. However, subtle DNAm changes over 12 months and technical variability may limit detection [3].

Exploratory pathways

GOmeth tests of time and the omnibus Timepoint × Profession interaction found no GO/KEGG terms at FDR < 0.05 [2].

Next step

Longer follow-up with exposure measurements, inflammatory markers and repeated methylation assessments could distinguish transient responses from sustained epigenetic change.

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[1] Hannum G et al. Molecular Cell. 2013;49:359-367. [2] Maksimovic J et al. Genome Biology. 2021;22:173.
[3] Higgins-Chen AT et al. Nature Aging. 2022;2:644-661. [4] Zhang Z et al. Aging Cell. 2024;23:e14071.

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